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"HOLISTIC CHILD CARE- Powered by Python and Renpy"

Department of Computer Science & Engineering (Data Science)

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Session 2025-26

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INTRODUCTION

- Many parents face significant challenges in accessing ***trustworthy, comprehensive, and culturally relevant guidance*** that supports both **effective parenting and holistic child development**.
- The lack of a unified and integrated digital platform makes it difficult for parents to monitor their child's health, developmental milestones, and overall well-being effectively.
- This gap often leads to parental uncertainty, confusion, and missed opportunities to provide timely support, resulting in suboptimal growth and weaker emotional connections with their children.
- **Addressing these challenges** with a comprehensive digital ***solution is essential to empower parents and promote the healthy development and safety of children.***

OBJECTIVE

- The objective is to develop a **unified, culturally relevant digital platform** providing trusted parenting guidance and tools to track child health, milestones, and emotional well-being, enabling confident and connected parenting.
- **Trusted Parenting Guidance** which provide reliable, culturally relevant parenting advice to support effective practices.
- **Health & Development Tracking** offers integrated tools for monitoring child health, vaccinations, and milestones.
- **Child Engagement & Safety** as it include interactive features that teach values, safety, and foster emotional well-being and also provides **Community Support** to build a connected parent community for sharing experiences and expert advice.

PROBLEM STATEMENT

- Parents often lack a single reliable and culturally relevant source to guide both effective parenting and holistic child development.
- The absence of integrated digital tools makes it challenging to monitor children's health, milestones, and emotional well-being.
- These gaps lead to parental uncertainty and missed opportunities for fostering optimal growth and strong emotional bonds in children.

NOVELTY OF PROJECT

- **Multi-lingual-support:** The platform is accessible in several Indian languages, making it usable for diverse families across regions.
- **Culturally-relevant-Indian-guidance:** Parenting tips and resources are tailored to Indian values, customs, and daily realities.
- **Integrated AI- powered gamified tools:** To engage children in mental health assessment through play and self-reflection.
- **Seamless integration with healthcare:** Enables real-time child health monitoring, personalized medical guidance, appointment scheduling, ensuring comprehensive support for child wellbeing.
- **Community Services:** Help kids to develop responsibility, and social skills by participating in group activities and peer support, and parents to connect-share experience in parenting groups or workshop.
- **Moral-Based Learning:** Teach values like kindness, honesty, and respect through stories, games, and real-life examples.

Literature Survey

Title	Year Published	Author(s)	Existing work	Pros	Cons
Evaluating Existing Mobile Apps for Toddler Parenting: Recommendations for Future App Developers	2024	Sunita, Poonam Malik	Evaluated 40 toddler-parenting apps from Google Play. Most were foreign (50%), in English (65%), and non-interactive (60%). Highlights the need for localized, interactive, and culturally relevant parenting tools.	Many toddler parenting apps available, Provide basic guidance via smartphones, Mostly in English, easy for urban users to access.	Not localized for Indian culture and languages, Mostly non-interactive, low engagement, focus on info, not personalized mental health, limited multi-language support.
Evaluating the Effectiveness of the Supportive Parenting App on Parental Outcomes: Randomized Controlled Trial	2023	Shefaly Shorey, Evelyn Law, Thilagamangai, Jancy Mathews, Siew Hoon Lim, Luming Shi, Jing Shi Chua, Ruochen Du, Yiong Huak Chan, Thiam Chye Tan, Cornelia Chee, Yap Seng Chong	Evidence-based parenting app for new parents providing guidance, interactive activities, and milestone tracking, mainly focused on infants and specific cultural contexts.	Includes activities to enhance parent-infant bonding, parents can monitor milestones and child development, developed with input from parents to improve usability, RCT shows improvements in parenting confidence and parent-child interaction.	Limited Cultural Adaptation, Restricted Scope, limited language Options, no Integration With Healthcare.
Beyond the Interface: Benchmarking Pediatric Mobile Health Applications for Monitoring Child Growth Using MARS	2025	M. Setyaningrum, N. E. Putri, R. Amalia, S. N. Hidayah, W. Lestari	Pediatric mHealth apps in Indonesia analyzed with MARS scale	Strong in app functionality and aesthetics, Improve parental awareness and care-seeking, Commercial apps often outperform non-commercial ones.	Weak in user engagement and info credibility, Few interactive features, Commercial focus may reduce equity and public interest.
I Become More Confident': Mother Use of Online Platform for Parenting Information	2020	Syifa Syarifah Alamiyah	Study of Indonesian mothers using online parenting platforms	Improves parenting knowledge, Boosts confidence, Facilitates peer support and community	Quality and reliability of information can vary, Some topics discussed are less relevant, Not all mothers access online platforms.
Parenting: There is an app for that. A systematic review of parenting interventions apps	2024	Oana Alexandra David, Ioana Alexandra Iuga, Ionela Simona Miron	Systematic review of 53 parenting apps (age 2–12)	Many apps free or low cost, Range of features (e.g. behavior tracking), Some use evidence-based techniques	Most apps lack personalization, Few clinical trials for effectiveness, Psychoeducation often missing.

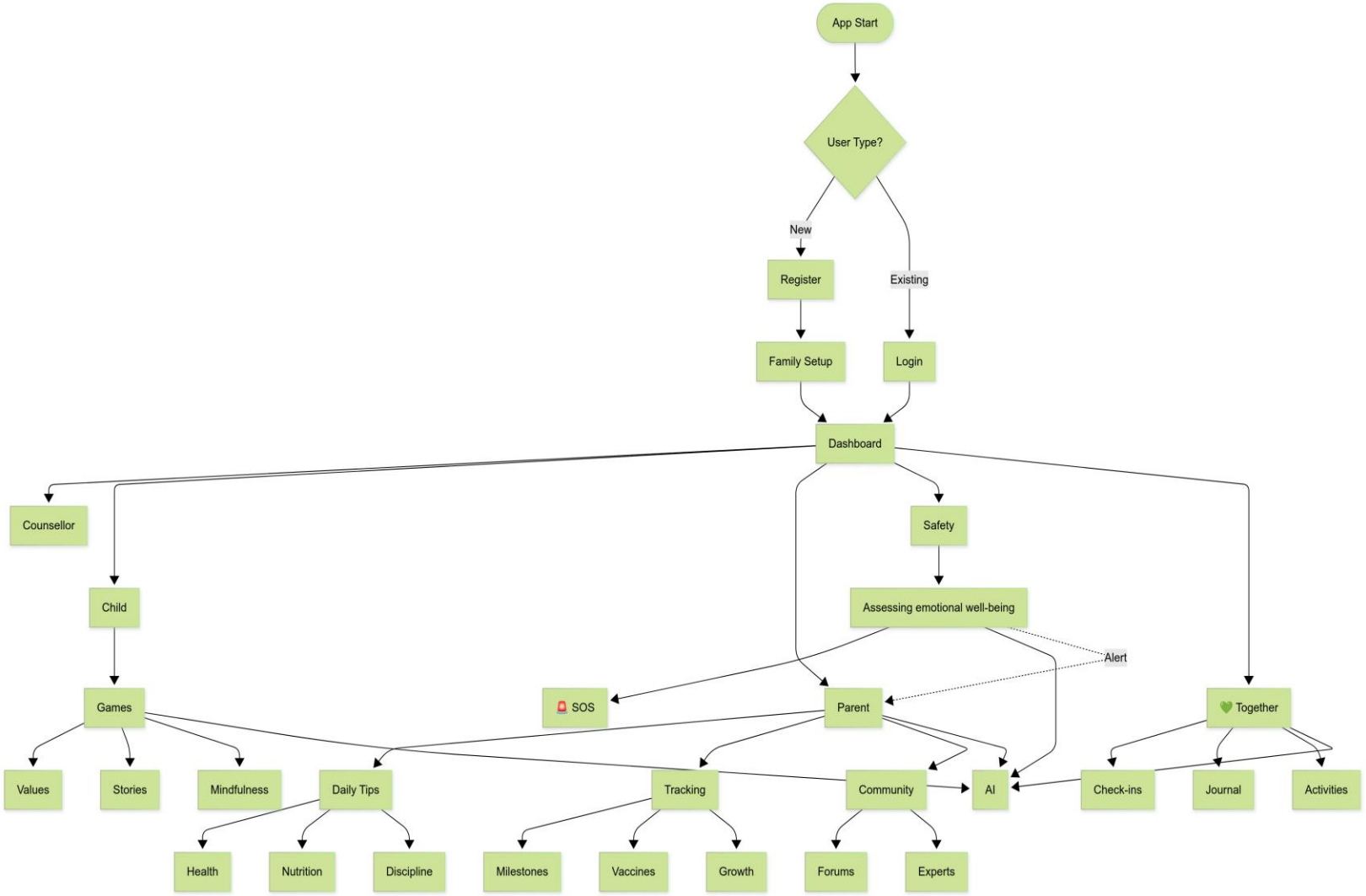
Title	Year Published	Author(s)	existing work/Keywords	Pros	Cons
Development of a Supportive Parenting App to Improve Parent and Infant Outcomes in the Perinatal Period.	2021	Shefaly Shorey, Thiam Chye Tan, thilagamangai, Jancy Mathews, Chun Yan Yu, Siew Hoon Lim.	Early-maternal health app(Andriod&IOS),adminportal-ASP.NET, C ; informational videos and audio clips + discussion forum.	Perinatal-period focus, co-designed with parents & healthcare staff, standardized information for parents, app run on both (IOS&Android).	excludes high-risk or complicated cases,technical issues, did not account for real-life and interpersonal challenges.
Exploring Maternal and Infant Health App	2024	Phillips JC, Alfano AR, Barfield LC, Cain L, Sadjadi M	mHealth app; Pregnancy and postpartum support, guidance for parents, self-care tools, health tracking, educational content.		
Parenting in the Digital era	2025	Arun B Nai	Digital Era, Parenting, Mental Health, Child development, Behavioural addiction	The paper provides recommendations for parents to navigate the digital era, focusing on the importance of parents being digitally and mentally health literate to guide their children. It emphasizes the need for an authoritative parenting approach, non-judgmental listening, and empowering children to make healthy	It lacks an empirical study, data, or experimental analysis, and as a result, does not have the structured weaknesses, limitations, or drawbacks typically found in academic research.
Parenting interventions to promote early child development in the first three years of life	2021	Joshua Jeong, Emily E. Franchett, Clariana V. Ramos de Oliveira, Karima Rehmani, Aisha K. Yousafzai	Parenting interventions, early child development, systematic review, meta-analysis, randomized controlled trials (RCTs)	This is the largest and most comprehensive global systematic review and meta-analysis on the topic. It found that parenting interventions were effective in improving a wide range of child development and parenting outcomes across low-, middle-, and high-income countries. The study also identified that interventions focusing on responsive caregiving had significantly greater effects	The study's limitations include variability and inadequate reporting of intervention content and implementation. No clear evidence that the interventions reduce the parental depressive symptoms. The authors noted that many of the included trials were not preregistered, which could lead to selective outcome reporting bias
The Effectiveness and Cost-Effectiveness of a Universal Digital Parenting Intervention Designed and Implemented During the COVID-19 Pandemic: Evidence From a Rapid-Implementation Randomized Controlled Trial Within a Cohort	2023	Melanie Palmer et al.	Parenting, digital application, randomized controlled trial, COVID-19 pandemic, mobile phone	The Parent Positive app was found to be cost-effective and effective in reducing child emotional problems. The findings suggest the app has potential for public health if implemented on a large scale.	The study did not find significant effects on child conduct problems. There was a high dropout rate in the intervention group, and the app was associated with increased parental

PROPOSED SOLUTION

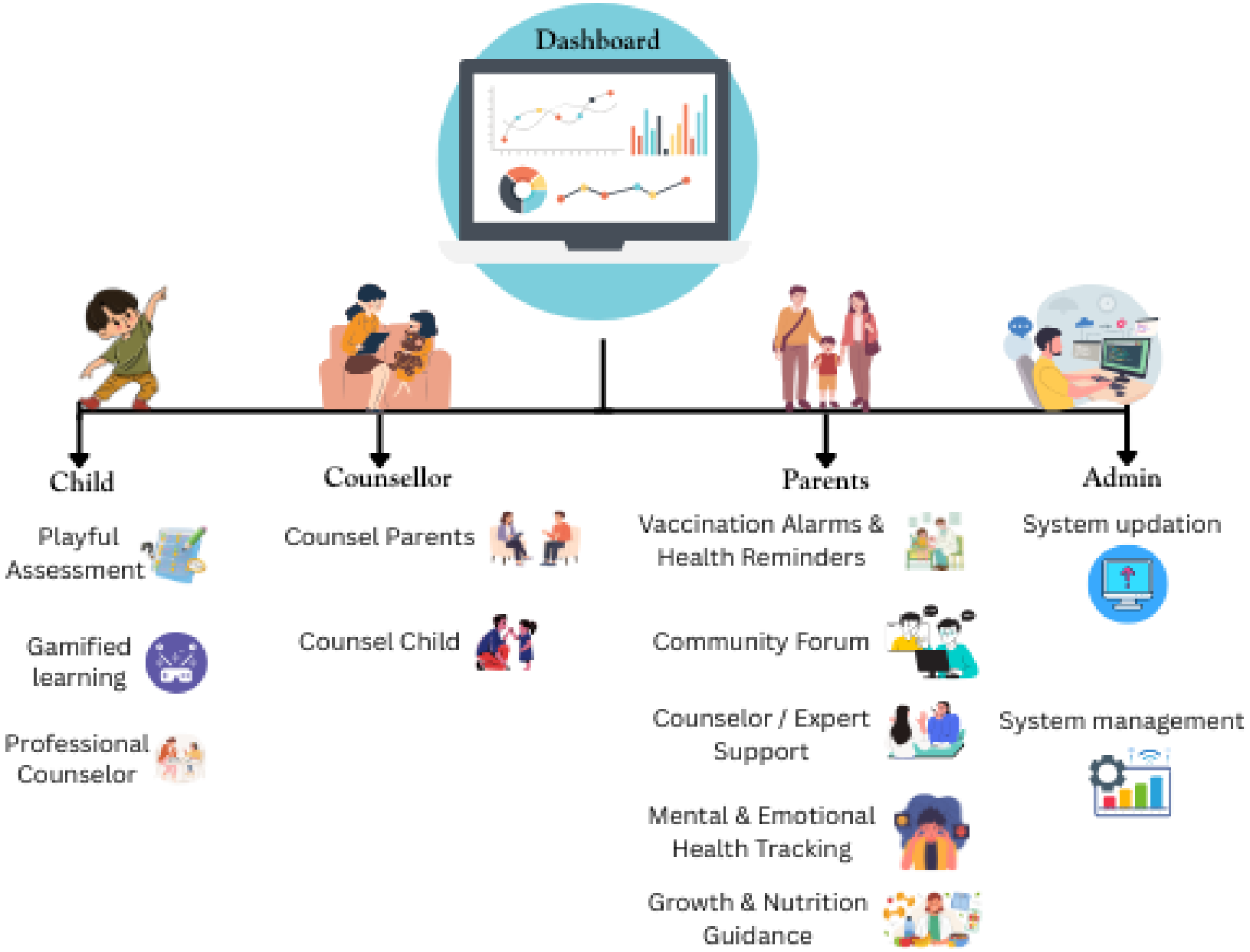
TECHNOLOGY USED

- Frontend: React
- Backend: Firebase and Database
- Game Engine: Ren'Py and Python

FLOWCHART



Methodology: The Users of our platform



The Child: The Learner

Assessments

Simple, age-appropriate quizzes to understand emotional, social, and behavioral well-being.

Games/Playful Activities

Interactive games designed to encourage learning and self-expression in a stress-free way.

Moral based learning cources

Short, story-driven lessons that teach empathy, kindness, honesty, and personal safety.

Our Counselor/Expert

Assessment Review

Access to reports generated by the platform to evaluate child's emotional or social needs.

One-to-One Consultation

Schedule and conduct online or in-person sessions with parents or children.

Our Root : The Parents

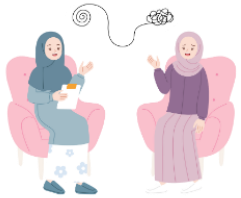
Vaccination Alarms & Health Reminders

Automatic alerts for important health check-ups and vaccination schedules.

Community Forum

Safe space to share experiences, ask questions, and learn from other parents.

Counselor / Expert Support



Easy access to psychologists, pediatricians, nutritionists, and parenting coaches.

Mental & Emotional Health Tracking

See reports from child's gamified assessments for stress, bullying, or behavioral changes.

Growth & Nutrition Guidance

Age-appropriate meal plans, recipes, and healthy routine suggestions.

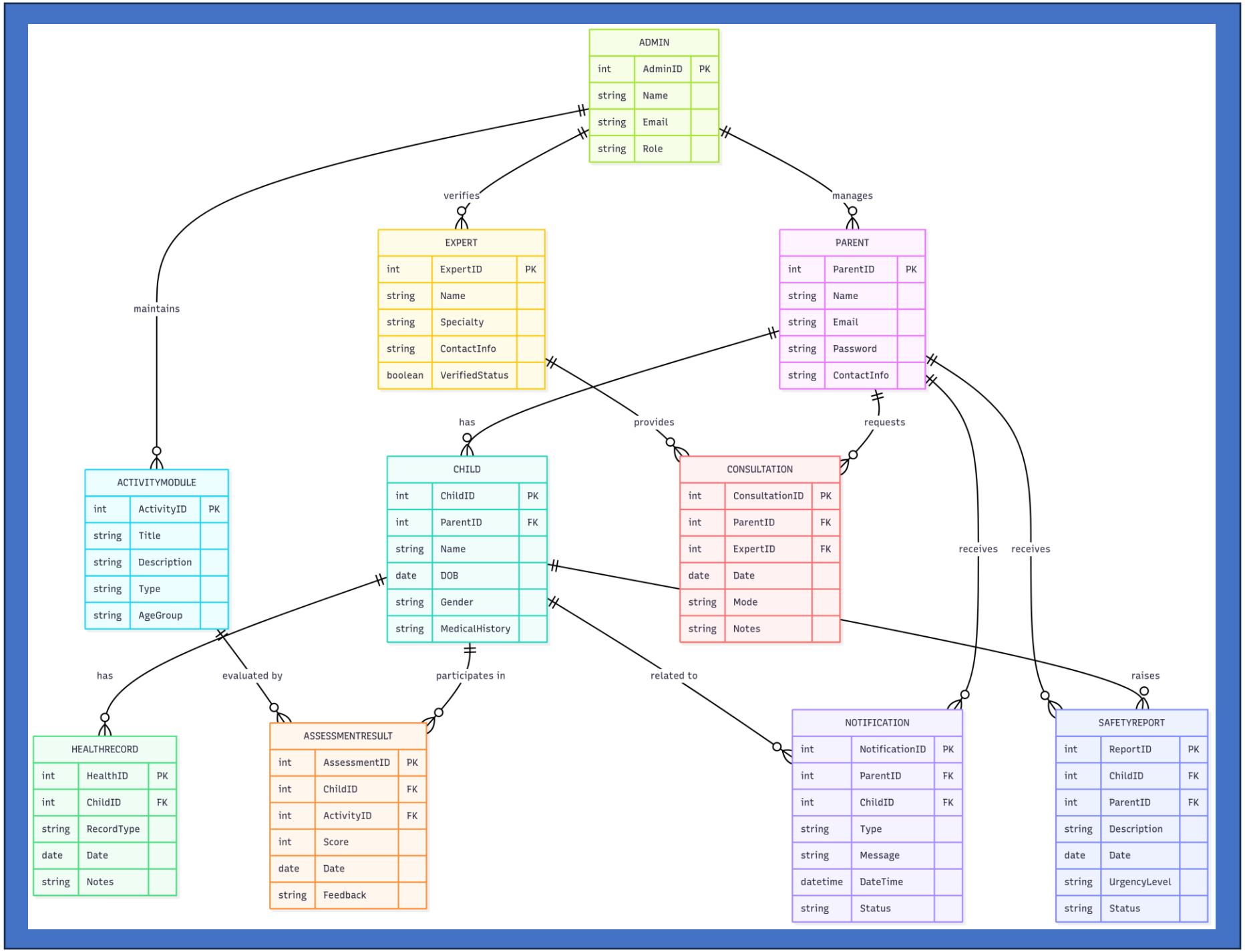
USE CASE

Use Case

A parent uses the platform to assess their child's mental health via interactive games.

- **Actors:** Parent, child and Expert/Counselor.
- **Process:** The parent logs in, the child plays a mental health assessment game, and the platform analyzes results to provide personalized parenting guidance.
- **Outcome:**
- The parent receives actionable recommendations to support the child's wellbeing.
- The child grows in a complete monitored and moral based environment.

ER DIAGRAM


ADMIN
int AdminID PK
string Name
string Email
string Role
verifies
manages
maintains
EXPERT
int ExpertID PK
string Name
string Specialty
string ContactInfo
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PARENT
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string Name
string Email
string Password
string ContactInfo
has
provides
requests
receives
receives
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string Description
string Type
string AgeGroup
CHILD
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date DOB
string Gender
string MedicalHistory
CONSULTATION
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string Mode
string Notes
has
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raises
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int ChildID FK
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string Message
datetime DateTime
string Status
SAFETYREPORT
int ReportID PK
int ChildID FK
int ParentID FK
string Description
date Date
string UrgencyLevel
string Status

FIGURES

Welcome to Holistic Child Care

Grow, Learn, Thrive Together

Powered by Python + Ren'Py

Holistic Child Care Python + Ren'Py

Interests | Wellbeing | Counsellors | Roadmap | Community

Unlock Your Child's Passions
Discover what makes them shine

Mindful Family, Happy Family
Take care of yourself and your loved ones

Daily Mood Check In

Parental Stress Relief

Art
Creative expression and visual arts
Age 3-12 Quiz

Unlock Your Child's Passions
Discover what makes them shine

Art
Creative expression and visual arts
Age 3-12 Quiz

Science
Explore and discover the world
Age 3-12 Quiz

Reading
Stories and language development
Age 3-12 Quiz

Mindful Family, Happy Family
Take care of yourself and your loved ones

Daily Mood Check In

Parental Stress Relief

Connect with Experts

Dr. Sarah Chen
Child Psychology
4.9 Available

Dr. Michael Torres
Family Therapy
4.8 Available

DER Dr. Emily Rodriguez
Parental Support
4.9 Available

Infants	Toddlers	Preschoolers	School Age
0-12 months	1-3 years	3-5 years	6-12 years
Bonding, feeding, and early development	Independence, language, and exploration	Learning readiness and social skills	Academic support and character building
Sleep Training	Potty Training	School Prep	Homework Help
Feeding	Tantrums	Social Skills	Activities
Milestones	Language	Creativity	Friendships
1,247 parents	2,156 parents	1,893 parents	3,472 parents
Join Community	Join Community	Join Community	Join Community

Recent Discussions
What parents are talking about
View All →

Sarah M. Sleep
Tips for bedtime routine with toddlers?
2h ago 23 replies

Mike T. Education
Starting kindergarten prep - what should I focus on?
4h ago 31 replies

Lisa K. Nutrition
Healthy snack ideas that kids actually eat
6h ago 45 replies

Upcoming Events
Join fellow parents
View All →

Virtual Playdate for 3-5 year olds 12/15
Tomorrow at 10 AM
12 joined Join Event

Parent Support Circle - Managing Screen Time 8/12
Friday at 7 PM
8 joined Join Event

Family Craft Workshop 18/20
Saturday at 2 PM
18 joined Join Event

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